

BL-20 Industrial Boiler

Authorized Technician Safety Reference

SIMULATED TRAINING DOCUMENT - NOT A MANUFACTURER MANUAL

1. Equipment Identification

Equipment code	BL-20
Equipment type	Simulated packaged hot-water boiler
Service	Training heat loop
Normal operating pressure	2.0-3.0 bar
High pressure limit	4.0 bar
Authorized service	Qualified boiler technicians only
Manual revision	Training Revision 1.0

System Overview

The BL-20 training unit contains a burner, circulation pump, flame sensor, water-level controls, pressure switches and safety interlocks.

SAFETY PREREQUISITES

Authorized technicians only. Shut down combustion. Isolate fuel and electrical energy and apply lockout/tagout to each source. Confirm zero pressure and full cooling. Wear heat-resistant, eye and hand PPE. Follow the site permit procedure.

2. Troubleshooting

Burner Flame Failure Lockout

SAFETY PREREQUISITES

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Observed symptoms

The burner stops, the flame-failure indicator is active, ignition does not continue, or the controller enters lockout.

Possible causes listed in this manual

Documented causes are contaminated or misaligned flame sensor, interrupted fuel supply, ignition electrode fault, combustion-air problem, or an open safety interlock.

Inspection guidance

Do not repeatedly reset the burner. An authorized technician must isolate all energy, confirm fuel isolation, inspect the flame sensor and ignition system, and test every safety interlock before restart.

Stop / refusal condition

If the safety prerequisites cannot be confirmed, stop. The manual does not authorize diagnosis or repair without isolation, zero-energy verification and required PPE.

3. Authorized Inspection Procedure

SAFETY PREREQUISITES

Authorized technicians only. Shut down combustion. Isolate fuel and electrical energy and apply lockout/tagout to each source. Confirm zero pressure and full cooling. Wear heat-resistant, eye and hand PPE. Follow the site permit procedure.

- Step 1. Record the controller fault code without resetting it.
- Step 2. Stop combustion and isolate fuel and electrical energy.
- Step 3. Verify zero pressure and safe temperature.
- Step 4. Inspect flame sensor cleanliness and position.
- Step 5. Inspect ignition electrode and combustion-air path.
- Step 6. Test safety interlocks using the approved site procedure.
- Step 7. A qualified supervisor must authorize restart.

Reference Parts

BL20-FS-01	Flame sensor
BL20-IGN-02	Ignition electrode
BL20-PRS-03	Pressure switch
BL20-GSK-11	Burner gasket kit

All work orders generated from this document require explicit supervisor review and approval before persistence or execution.